

Computational Modeling of Dendritic-Somatic Interactions Driven by Electric Field Polarization in Pyramidal Neurons*

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Abstract—Electrical stimulation has been widely used in the treatment of neurological disorders. At the single-neuron level, its efficacy depends on the polarization effects of the induced electric field (EF) on neuronal firing patterns. However, the interactions between dendritic firing and somatic action potentials under EF exposure remain poorly understood. In this study, we develop a pyramidal neuron model to investigate the effects of EF polarization on dendritic-somatic interactions. Simulation results reveal that increasing EF magnitude induces distinct firing trends in the apical and basal dendrites, which ultimately influence the generation of action potentials in the soma. Bifurcation analysis elucidates their underlying dynamics, including subcritical Hopf and saddle-node on invariant circle bifurcations, which delineate the parameter boundaries for dendritic oscillations. Notably, the orientation of the limit cycle branches in the bifurcation diagram reflects the polarization effects of EF on dendrites.

Clinical relevance—These findings offer novel insights into the microscopic biophysical mechanisms of EF-regulated neuronal firing patterns, thereby providing a theoretical framework for optimizing parameters in electrical stimulation protocols.

Keywords—Computational modeling, Electric Field, Pyramidal Neuron, Bifurcation Analysis, Electrical Stimulation

I. INTRODUCTION

Electrical stimulation is a neuromodulation technique employing invasive or non-invasive methods to modulate neuronal firing activity in the central and peripheral nervous systems [1]. With the development of neuroscience and biomedical engineering, electrical stimulation has become a rapidly evolving field in rehabilitation medicine. For example, deep brain stimulation (DBS) has been validated as an effective therapy for Parkinson's disease. By precisely implanting electrodes into the subthalamic nucleus, DBS delivers optimized electrical pulses to targeted regions, stimulating neural activity to alleviate motor symptoms [2]. Similarly, transcranial direct current stimulation (tDCS) modulates cortical excitability and synaptic plasticity via weak currents applied to the scalp, commonly used to treat intractable epilepsy [3]. Despite their clinical success, the mechanisms by which these techniques modulate neuronal firing patterns remain poorly understood. This gap in knowledge, combined with

inter-individual variability, hinders the precise optimization of stimulation parameters [4].

Although DBS and tDCS differ in methodology, recent neurophysiological studies suggest that both rely on similar biophysical mechanisms involving electric field (EF). A study on tDCS demonstrated that weak applied currents can penetrate through the skull to induce EFs in the cortex, with magnitude proportional to the currents [5]. These tDCS-induced EFs modulate the membrane potentials of cortical neurons via depolarization or hyperpolarization, affecting regional brain activity [6]. Likewise, the stimulation pulses delivered by DBS also generate EFs around implanted electrodes, regulating the firing patterns of neurons by altering excitation-inhibition balance at specific synapses [7]. Thus, at the single-neuron level, the therapeutic efficacy of both DBS and tDCS partly depends on the polarization effects of induced EFs.

Pyramidal neurons, as key targets of electrical stimulation, exhibit distinct firing activities in their dendrites and soma. However, the interactions between dendritic firing and somatic action potentials under EF polarization remain poorly understood, highlighting the need for computational modeling to uncover their complex dynamics [8]. In this study, we develop a pyramidal neuron model to investigate dendritic-somatic interactions driven by EF polarization at the single-neuron level. This model includes apical dendrite, soma, and basal dendrite, which generate Ca^{2+} -, Na^{+} -, and N-methyl-D-aspartate (NMDA) receptor-mediated firing activities, respectively. Simulation results indicate that as the EF magnitude increases, firing activities in the apical and basal dendrites exhibit distinct trends, which, in turn, influence the generation of somatic action potentials. Moreover, bifurcation analysis elucidates the dynamics underlying EF-induced firing pattern transitions, delineating the parameter boundaries for dendritic firing activities. These findings deepen our understanding of how EF polarization governs the interactions between dendritic and somatic firing activities, offering insights for optimizing parameters in electrical stimulation protocols.

II. MATERIALS AND METHODS

A. Pyramidal Neuron Model under Electric Field Exposure

In our previous work [9], we developed a three-compartment pyramidal neuron model to replicate multi-timescale compound oscillations observed in epilepticform discharges. In this study, the model is adopted and subjected to an external EF to investigate the effects of EF polarization on firing patterns, as shown in Fig. 1. The three-compartment

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model is chosen based on the following considerations: (1) pyramidal neurons are the fundamental excitatory units in the neocortex and hippocampus, primary targets of tDCS; (2) the elongated dendritic morphology of pyramidal neurons, comprising apical and basal dendrites, renders them highly sensitive to EF polarization; (3) distinct firing activities in dendrites, interacting with somatic action potentials, enable diverse firing patterns, ideal for studying EF-induced firing pattern transitions.

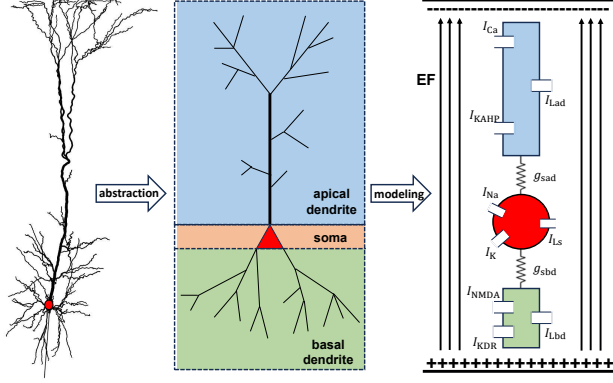


Fig. 1. Pyramidal neuron in the cortex and its three-compartment model, consisting of apical dendrite, soma, and basal dendrite, which capture realistic dendritic morphology. The model is subjected to an external electric field, with its positive direction defined from basal to apical dendrites, parallel to the axis of the neuron model.

B. Model Implementation

As depicted in Fig. 1, the model consists of apical dendrite, soma, and basal dendrite compartments, interconnected by axial conductances g_{sad} and g_{sbd} . The distances between the soma and the apical and basal dendrite are L_{sad} and L_{sbd} , respectively. Each compartment incorporates active depolarizing and hyperpolarizing ion currents driving firing activity. Specifically, the apical dendrite contains the voltage-gated calcium current I_{Ca} and the calcium-dependent afterhyperpolarization (AHP) potassium current I_{KAHP} , responsible for dendritic Ca^{2+} firing. The soma contains voltage-dependent sodium and potassium currents, I_{Na} and I_K , driving Na^+ action potentials. The basal dendrite contains the NMDA receptor-mediated current I_{NMDA} and the delayed rectifier potassium current I_{KDR} , responsible for dendritic NMDA firing. Additionally, each compartment contains passive leakage currents, denoted I_{Lad} , I_{Ls} , and I_{Lbd} . Under an uniform EF of magnitude E , the membrane potentials V_{ad} , V_s , and V_{bd} of the apical dendrite, soma, and basal dendrite, along with their gating variables, are governed by the following system:

$$C \frac{dV_{ad}}{dt} = -I_{Ca} - I_{KAHP} - I_{Lad} + \frac{g_{sad}}{A_{ad}}(V_s - V_{ad} + E \cdot L_{sad}), \quad (1)$$

$$\frac{dh_{Ca}}{dt} = \frac{h_{Ca,\infty}(V_{ad}) - h_{Ca}}{\tau_{h_{Ca}}}, \quad (2)$$

$$\frac{d[Ca^{2+}]i}{dt} = -0.13I_{Ca} - \frac{[Ca^{2+}]i}{\tau_{[Ca]}}, \quad (3)$$

$$\frac{dq}{dt} = \frac{q_{\infty}([Ca^{2+}]i) - q}{\tau_q}, \quad (4)$$

$$C \frac{dV_s}{dt} = -I_{Na} - I_K - I_{Ls} + \frac{g_{sad}}{A_s}(V_{ad} - V_s - E \cdot L_{sad}) + \frac{g_{sbd}}{A_s}(V_{bd} - V_s + E \cdot L_{sbd}), \quad (5)$$

$$\frac{dw}{dt} = 0.15 \frac{w_{\infty}(V_s) - w}{\tau_w(V_s)}, \quad (6)$$

$$C \frac{dV_{bd}}{dt} = -I_{NMDA} - I_{KDR} - I_{Lbd} + \frac{g_{sbd}}{A_{bd}}(V_s - V_{bd} - E \cdot L_{sbd}), \quad (7)$$

$$\frac{dn}{dt} = \alpha_n(V_{bd}) - n(\alpha_n(V_{bd}) + \beta_n(V_{bd})), \quad (8)$$

where A_{ad} , A_s , and A_{bd} are the membrane areas of the apical dendrite, soma, and basal dendrite, respectively, and C is the membrane capacitance. The ion currents are modeled using the Hodgkin-Huxley formalism as follows:

$$I_{Ca} = g_{Ca} \cdot m_{Ca,\infty}^2(V_{ad}) \cdot h_{Ca} \cdot (V_{ad} - E_{Ca}), \quad (9)$$

$$I_{KAHP} = g_{KAHP} \cdot q \cdot (V_{ad} - E_K), \quad (10)$$

$$I_{Lad} = g_{Lad} \cdot (V_{ad} - E_L), \quad (11)$$

$$I_{Na} = g_{Na} \cdot m_{Na,\infty}(V_s) \cdot (V_s - E_{Na}), \quad (12)$$

$$I_K = g_K \cdot w \cdot (V_s - E_K), \quad (13)$$

$$I_{Ls} = g_{Ls} \cdot (V_s - E_L), \quad (14)$$

$$I_{NMDA} = g_{NMDA} \cdot B(V_{bd}) \cdot (V_{bd} - E_{NMDA}), \quad (15)$$

$$I_{KDR} = g_{KDR} \cdot n^4 \cdot (V_{bd} - E_K), \quad (16)$$

$$I_{Lbd} = g_{Lbd} \cdot (V_{bd} - E_L). \quad (17)$$

In this system, (1)–(4) describe the dynamics of the apical dendrite, (5)–(6) describe the soma, and (7)–(8) describe the basal dendrite. Unless otherwise specified, the values for maximum conductance densities, reverse potentials, and other parameters are taken from Table I.

TABLE I
PARAMETER VALUES FOR THE THREE-COMPARTMENT MODEL

Parameters	Values	Parameters	Values
g_{Ca}	3 mS/cm ²	E_{Ca}	20 mV
g_{KAHP}	0.03 mS/cm ²	E_K	-80 mV
g_{Na}	15 mS/cm ²	E_{Na}	50 mV
g_K	32 mS/cm ²	E_{NMDA}	0 mV
g_{NMDA}	1.7 mS/cm ²	E_L	-70 mV
g_{KDR}	25 mS/cm ²	$\tau_{h_{Ca}}$	157 ms
g_{Lad}	0.05 mS/cm ²	$\tau_{[Ca]}$	13 ms
g_{Ls}	1.47 mS/cm ²	τ_q	300 ms
g_{Lbd}	0.05 mS/cm ²	A_{ad}	3.58×10^{-4} cm ²
g_{sad}	5.70×10^{-6} mS/cm ²	A_s	2.46×10^{-5} cm ²
g_{sbd}	3.13×10^{-5} mS/cm ²	A_{bd}	1.01×10^{-4} cm ²
L_{sad}	1100 μ m	C	1.2 μ F/cm ²
L_{sbd}	200 μ m		

It should be noted that (1), (5), and (7) differ from the original model in [9] due to the incorporation of EF terms. The derivation of these equations is detailed in Appendix. The physiological significance of all gating variables and the definitions of associated gating functions are also provided

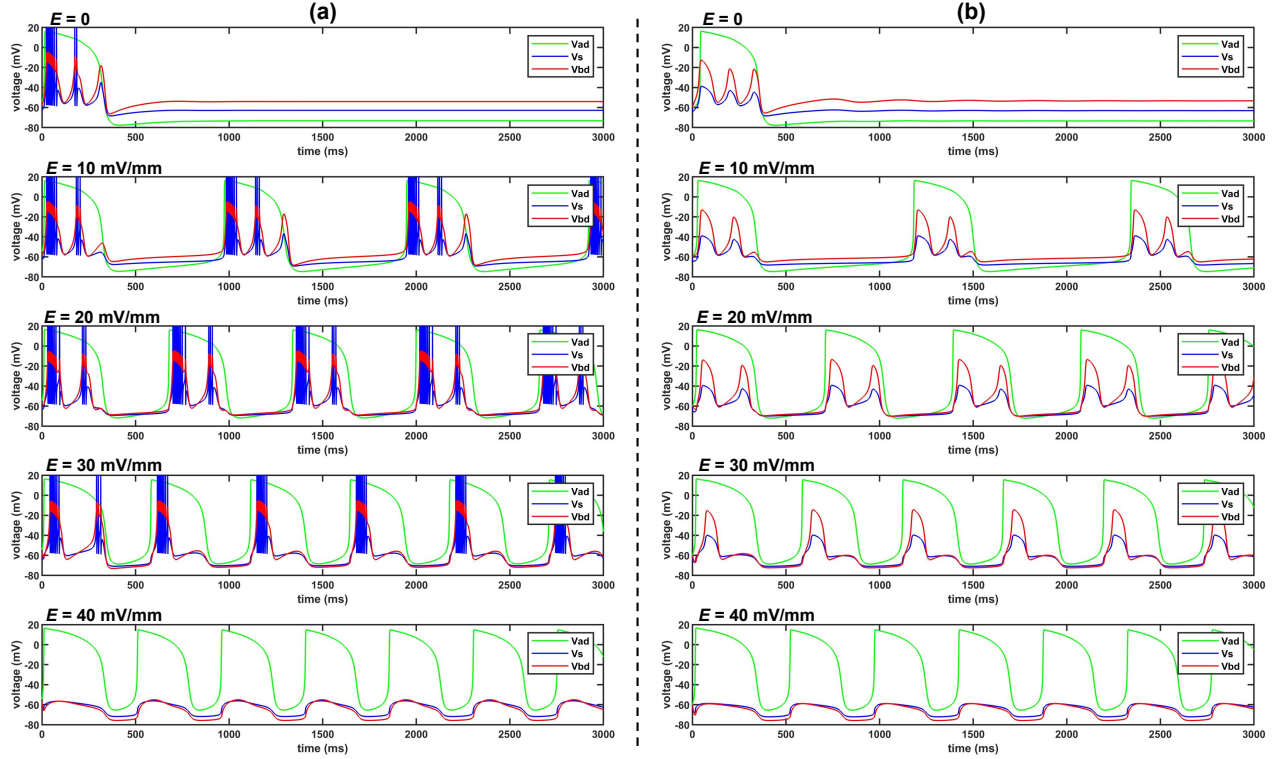


Fig. 2. (a) Simulation results of the membrane potentials for varying electric field magnitude E . The model converges to a stable resting state when $E = 0$. The green curve represents the membrane potential of the apical dendrite (V_{ad}), the blue curve the soma (V_s), and the red curve the basal dendrite (V_{bd}). (b) Simulation results with blocked sodium channels ($g_{Na} = 0$). Notably, the basal dendrite exhibits bursting in (a), while oscillatory firing in (b).

in Appendix. The entire model is implemented in XPPAUT and integrated numerically using 4th order Runge-Kutta algorithm with an integration time step of 0.05 ms. Moreover, bifurcation analysis is conducted following the standard procedures in XPPAUT [10].

III. RESULTS

A. Apical and Basal Dendrites Exhibit Distinct Firing Trends under EF Polarization

In this section, we examine the firing patterns of the model under varying electric field magnitude E . As action potentials are critical for information communication in the nervous system, we first focus on the response of somatic Na^+ action potentials to changes in E . Simulation results, shown in Fig. 2(a), reveal that as E increases from 0 to 10 mV/mm, the neuron model transitions from a resting state to a periodic Ca^{2+} -NMDA- Na^+ compound oscillation mode. This transition occurs in two steps: equation (1) indicates that, under EF polarization, a positive E is equivalent to a depolarizing current for the apical dendrite, triggering Ca^{2+} oscillations. These apical dendritic oscillations subsequently induce NMDA firing in the basal dendrite and Na^+ action potentials in the soma, as detail in [9]. Moreover, the frequency of Ca^{2+} oscillations increases monotonically with E , whereas the frequencies of NMDA and Na^+ firings increase initially and then decrease. Notably, when E reaches 40 mV/mm, Na^+ action potentials disappear, indicating that Na^+ firing only

occurs within a specific range of electric field magnitude.

Our previous work [9] have demonstrated that Na^+ action potentials depend on both Ca^{2+} oscillations in the apical dendrite and NMDA firing in the basal dendrite. To explain the disappearance of Na^+ action potentials, we set $g_{Na} = 0$ to isolate the dendritic firing trends under varying E . As shown in Fig. 2(b), the frequency of Ca^{2+} oscillation still increases with E . However, during Ca^{2+} oscillations, the basal dendrite exhibits NMDA oscillations (in contrast to the bursting mode observed in Fig. 2(a)), whose frequency decreases as E further increases. This suggests that the disappearance of Na^+ action potentials in Fig. 2(a) can be attributed to the suppression of NMDA firing in the basal dendrite.

Equation (7) indicates that, under EF polarization, a positive E is equivalent to a hyperpolarizing current for the basal dendrite. At lower E , the hyperpolarization is relatively weak, and the axial current generated by Ca^{2+} firing is sufficient to induce NMDA oscillations in the basal dendrite. As E increases, the aforementioned hyperpolarization becomes dominant, and when E exceeds a critical value, Ca^{2+} oscillations fail to trigger NMDA firing. Thus, under EF polarization, the firing pattern of the basal dendrite is regulated by two opposing mechanisms: the excitatory axial current from internal Ca^{2+} firing and the inhibitory hyperpolarization from the external EF. Changes in E alter this excitation-inhibition balance, resulting in the observed firing pattern transitions.

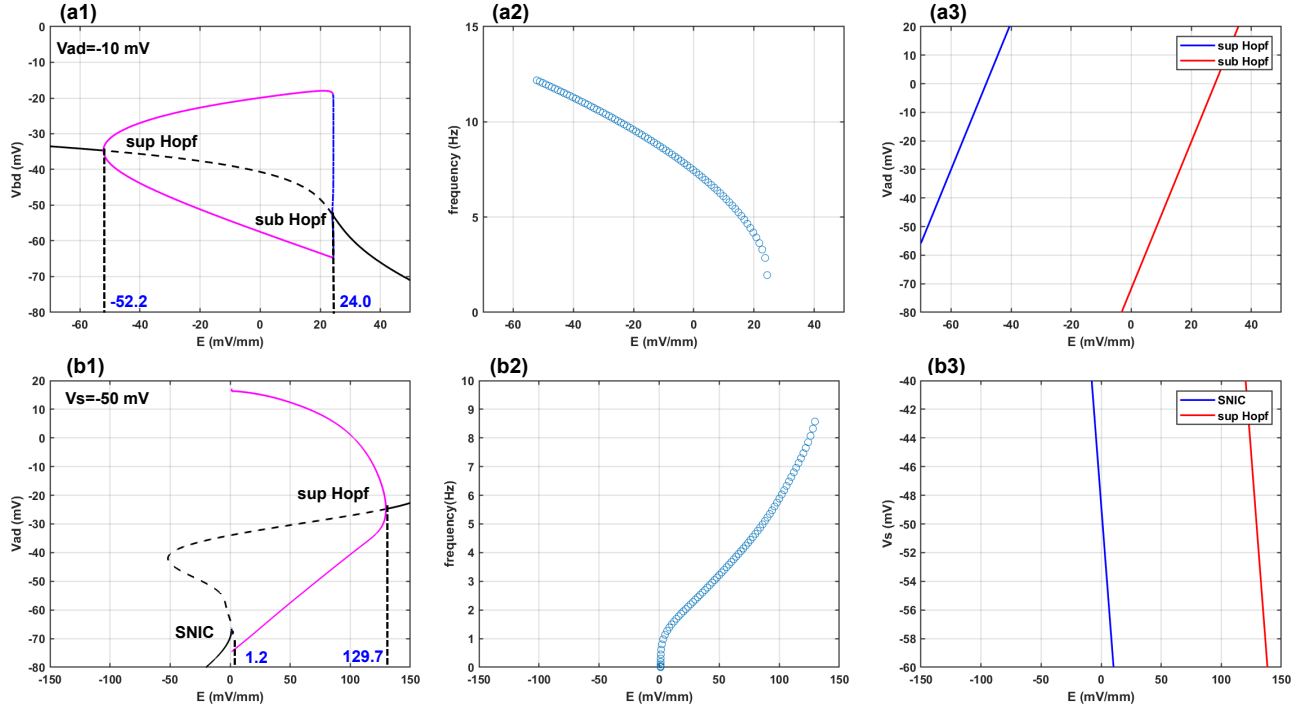


Fig. 3. **(a)** Bifurcation analysis of the NMDA oscillator ((5)–(8)). (a1) One-parameter bifurcation diagram of V_{bd} with respect to E , with $V_{ad} = -10$ mV. The black curve represents equilibrium points of V_{bd} (solid: stable; dashed: unstable). The magenta curve represents stable limit cycles, and the blue dashed curve represents unstable limit cycles. (a2) NMDA oscillation frequency as a function of E ($f-E$ curve). (a3) Two-parameter bifurcation diagram of V_{bd} in the (E, V_{ad}) parameter space, illustrating the dependence of bifurcation points from (a1) on V_{ad} . **(b)** Bifurcation analysis of the Ca^{2+} oscillator ((1)–(4)). (SNIC: Saddle-Node on Invariant Circle bifurcation; *sub Hopf*: subcritical Hopf bifurcation; *sup Hopf*: supercritical Hopf bifurcation.)

In summary, under EF polarization, the apical and basal dendrites exhibit distinct firing trends as E increases, which influences the generation of somatic action potentials. From the perspective of electrical stimulation, the magnitude of induced EF must be precisely controlled to avoid excessive suppression of somatic action potentials, which could alter cortical and hippocampal rhythms [11].

B. Dynamical Mechanisms of EF-Induced Firing Pattern Transitions in the Apical and Basal Dendrites

In this section, we elucidate the dynamical mechanisms underlying the EF-induced firing pattern transitions observed in Fig. 2(b). To facilitate the analysis, we decompose the neuron model into two subsystems: the NMDA oscillator ((5)–(8)) and the Ca^{2+} oscillator ((1)–(4)).

For the NMDA oscillator, the key parameters are V_{ad} and E . To investigate the transition of basal dendrite firing patterns with varying E , we fix $V_{ad} = -10$ mV and perform a one-parameter bifurcation analysis of V_{bd} with respect to E . The resulting bifurcation diagram, shown in Fig. 3(a1), reveals the presence of stable limit cycles (i.e., NMDA oscillations) within the range $E \in (-52.2, 24.0)$ mV/mm. At $E = 24.0$ mV/mm, the limit cycle is terminated via a subcritical Hopf bifurcation, while at $E = -52.2$ mV/mm, it is terminated via a supercritical Hopf bifurcation. The corresponding $f-E$ curve (Fig. 3(a2)) shows that the NMDA oscillation frequency decreases as E increases. Furthermore, the two-parameter bifurcation diagram in the (E, V_{ad}) space

(Fig. 3(a3)) indicates that as V_{ad} increases, both Hopf bifurcation points shift to higher E values. Due to $V_{ad} \in [-80, 20]$ mV during Ca^{2+} oscillations, Fig. 3(a3) provides an intuitive explanation for the disappearance of NMDA oscillations at $E = 40$ mV/mm in Fig. 2(b). At this value, E consistently remains to the right of the subcritical Hopf bifurcation points.

Similarly, for the Ca^{2+} oscillator, the key parameters are V_s and E . To study the transition of apical dendritic firing patterns with varying E , we fix $V_s = -50$ mV and perform a one-parameter bifurcation analysis of V_{ad} with respect to E . As shown in Fig. 3(b1), when E exceeds 1.2 mV/mm, V_{ad} transitions from an equilibrium state to a limit cycle via a Saddle-Node on Invariant Circle (SNIC) bifurcation, resulting in Ca^{2+} oscillations. As E increases beyond 129.7 mV/mm, the limit cycle terminates via a subcritical Hopf bifurcation, leading V_{ad} into a depolarized block state [12]. The corresponding $f-E$ curve (Fig. 3(b2)) indicates a continuous increase in oscillation frequency as E moves away from the SNIC point. The two-parameter bifurcation diagram in the (E, V_s) space (Fig. 3(b3)) reveals that both bifurcation points shift to lower E values with increasing V_s , though this dependency is relatively weak. This explains why Ca^{2+} oscillations consistently occur in Fig. 2 as long as E does not exceed the subcritical Hopf bifurcation point.

In summary, using bifurcation analysis, we elucidates the dynamics driving the firing patterns transition in the apical and basal dendrites under EF polarization. The bifurcation diagrams in Fig. 3(a1) and Fig. 3(b1) highlight opposite

orientations of the limit cycle branches. In Fig. 3(a1), the right-opening structure is bounded by a subcritical Hopf bifurcation on the right, corresponding to the hyperpolarization effect of positive E on the basal dendrite. Conversely, in Fig. 3(b1), the left-opening structure is bounded by a SNIC bifurcation on the left, corresponding to the depolarization effect of positive E on the apical dendrite.

IV. DISCUSSION AND CONCLUSIONS

Electrical stimulation techniques are extensively applied in treating neurological and psychiatric disorders. Optimizing stimulation parameters requires a deeper understanding of their underlying biophysical mechanisms. At the single-neuron level, the efficacy of electrical stimulation depends on the polarization effects of stimulus-induced EFs on neuronal activity. Studies indicate that these EFs can significantly modulate the firing patterns of cortical pyramidal neurons [13]. Given the complex dynamical interactions between dendritic firing and somatic action potentials under EF polarization, computational modeling offers valuable insights into this complexity. Accordingly, this study developed a three-compartment pyramidal neuron model to explore the effects of EF polarization on neuronal firing patterns. The main findings of this study are summarized below.

First, simulation results in Fig. 2(a) reveal that the frequency of somatic Na^+ action potentials does not increase monotonically with EF magnitude; instead, these Na^+ action potentials diminish and vanish at $E = 40$ mV/mm. This phenomenon arises from the role of dendritic plateau potentials, which underlie somatic action potentials. Simulation results in Fig. 2(b) indicate that the frequencies of apical dendritic Ca^{2+} oscillations and basal dendritic NMDA oscillations exhibit opposing trends as EF magnitude increases. This is because a positive E depolarizes the apical dendrite but hyperpolarizes the basal dendrite. At high E , hyperpolarization suppresses NMDA firing in the basal dendrite, reducing plateau potentials and leading to the cessation of somatic action potentials. These findings highlight the importance of precise EF intensity control in electrical stimulation. For disorders like depression, characterized by reduced neuronal excitability, stimulation intensity must be carefully controlled to avoid excessive suppression of neuronal activity [14]. Conversely, for disorders like epilepsy, with heightened neuronal excitability, EF-induced hyperpolarization may be leveraged to suppress epileptiform discharges [15].

Second, bifurcation analysis provided a mechanistic explanation for the transitions in dendritic firing patterns. As shown in Fig. 3(a1) and (b1), the limit cycle branches in the bifurcation diagrams exhibit opposite orientations, reflecting the contrasting polarization effects of EF on basal and apical dendrites. Specifically, the limit cycle branch in Fig. 3(a1) terminates at a subcritical Hopf bifurcation, corresponding to the abrupt disappearance of NMDA oscillations as E increases. In contrast, the limit cycle branch in Fig. 3(b1) is bounded by a SNIC bifurcation, corresponding to the gradual emergence of Ca^{2+} oscillations with increasing E . Furthermore, the bifurcation boundaries ($E < 24.0$ mV/mm

for NMDA oscillations; $E < 129.7$ mV/mm for Ca^{2+} oscillations) provide a theoretical framework for optimizing stimulation parameters, ensuring therapeutic efficacy without suppressing critical spikes or inducing depolarization block.

In conclusion, this study investigates dendritic-somatic interactions driven by EF polarization and elucidates the mechanisms underlying EF-regulated neuronal firing pattern transitions in a pyramidal neuron model. It should be noted that this study omits network-level interactions, which could potentially modify bifurcation structures and polarization patterns. Future research will extend the model to cortical network simulations and validate the findings against clinical data. Additionally, we will consider the orientation of the external EF relative to the neuron's axis and incorporate more complex dendritic morphologies to advance our understanding of EF-regulated neuronal behavior in more physiologically realistic settings [16].

APPENDIX

Here, we present the derivation of (1), (5), and (7) in the proposed model. In the absence of an external electric field, the membrane potentials V_{ad} , V_s , and V_{bd} of the apical dendrite, soma, and basal dendrite are governed by the following equations:

$$C \frac{dV_{ad}}{dt} = -I_{Ca} - I_{KAHP} - I_{Lad} + \frac{g_{sad}}{A_{ad}}(V_s - V_{ad}), \quad (A1)$$

$$C \frac{dV_s}{dt} = -I_{Na} - I_K - I_{Ls} + \frac{g_{sad}}{A_s}(V_{ad} - V_s) + \frac{g_{sbd}}{A_s}(V_{bd} - V_s), \quad (A2)$$

$$C \frac{dV_{bd}}{dt} = -I_{NMDA} - I_{KDR} - I_{Lbd} + \frac{g_{sbd}}{A_{bd}}(V_s - V_{bd}). \quad (A3)$$

Under external electric field polarization, the extracellular potentials $V_{ad,ex}$, $V_{s,ex}$, and $V_{bd,ex}$ at the three compartments are not equal. According to [17], (A1)–(A3) should be modified to account for the electric field exposure as follows:

$$C \frac{dV_{ad}}{dt} = -I_{Ca} - I_{KAHP} - I_{Lad} + \frac{g_{sad}}{A_{ad}}(V_{s,in} - V_{ad,in}), \quad (A4)$$

$$C \frac{dV_s}{dt} = -I_{Na} - I_K - I_{Ls} + \frac{g_{sad}}{A_s}(V_{ad,in} - V_{s,in}) + \frac{g_{sbd}}{A_s}(V_{bd,in} - V_{s,in}), \quad (A5)$$

$$C \frac{dV_{bd}}{dt} = -I_{NMDA} - I_{KDR} - I_{Lbd} + \frac{g_{sbd}}{A_{bd}}(V_{s,in} - V_{bd,in}), \quad (A6)$$

where $V_{ad,in}$, $V_{s,in}$, and $V_{bd,in}$ denote the intracellular potentials of the three compartments. Using the definition of transmembrane potential, $V = V_{in} - V_{ex}$, we rewrite (A4)–(A6) as:

$$C \frac{dV_{ad}}{dt} = -I_{Ca} - I_{KAHP} - I_{Lad} + \frac{g_{sad}}{A_{ad}}(V_s - V_{ad} + V_{s,ex} - V_{ad,ex}), \quad (A7)$$

$$C \frac{dV_s}{dt} = -I_{Na} - I_K - I_{Ls} + \frac{g_{sad}}{A_s}(V_{ad} - V_s + V_{ad,ex} - V_{s,ex}) + \frac{g_{sbd}}{A_s}(V_{bd} - V_s + V_{bd,ex} - V_{s,ex}), \quad (A8)$$

$$C \frac{dV_{bd}}{dt} = -I_{NMDA} - I_{KDR} - I_{Lbd} + \frac{g_{sbd}}{A_{bd}}(V_s - V_{bd} + V_{s,ex} - V_{bd,ex}). \quad (A9)$$

For an electric field of magnitude E , the extracellular potential differences are given by $V_{s,ex} - V_{ad,ex} = E \cdot L_{sad}$ and $V_{s,ex} - V_{bd,ex} = E \cdot L_{sbd}$. Substituting these into (A7)–(A9) yields the final form of the model:

$$C \frac{dV_{ad}}{dt} = -I_{Ca} - I_{KAHP} - I_{Lad} + \frac{g_{sad}}{A_{ad}}(V_s - V_{ad} + E \cdot L_{sad}), \quad (A10)$$

$$C \frac{dV_s}{dt} = -I_{Na} - I_K - I_{Ls} + \frac{g_{sad}}{A_s}(V_{ad} - V_s - E \cdot L_{sad}) + \frac{g_{sbd}}{A_s}(V_{bd} - V_s + E \cdot L_{sbd}), \quad (A11)$$

$$C \frac{dV_{bd}}{dt} = -I_{NMDA} - I_{KDR} - I_{Lbd} + \frac{g_{sbd}}{A_{bd}}(V_s - V_{bd} - E \cdot L_{sbd}), \quad (A12)$$

Next, the physiological significance of all gating variables and the definitions of associated gating functions are provided below. In the apical dendrite, m_{Ca} and h_{Ca} represent the activation and inactivation gating variables of I_{Ca} , respectively. Their steady-state functions, $m_{Ca,\infty}(V_{ad})$ and $h_{Ca,\infty}(V_{ad})$, are given in Boltzmann form as follows:

$$m_{Ca,\infty}(V_{ad}) = 1/(1 + \exp(-(V_{ad} + 20)/16)),$$

$$h_{Ca,\infty}(V_{ad}) = 1/(1 + \exp((V_{ad} + 40)/7)).$$

For the activation gating variables q of I_{KAHP} , its steady-state function $q_{\infty}([Ca^{2+}]_i)$ is as follows:

$$q_{\infty}([Ca^{2+}]_i) = \frac{1}{1 + 81/[Ca^{2+}]_i^4},$$

where $[Ca^{2+}]_i$ is dimensionless with time constant $\tau_{[Ca]}$ and is proportional to the Ca^{2+} concentration in a thin internal cylindrical shell adjacent to the membrane [18].

In the soma, m_{Na} and w are activation gating variables of I_{Na} and I_K , respectively. The associated steady-state functions and time constant are defined as follows:

$$m_{Na,\infty}(V_s) = 0.5(1 + \tanh((V_s + 1.2)/18)),$$

$$w_{\infty}(V_s) = 0.5(1 + \tanh(V_s/10)),$$

$$\tau_w(V_s) = 1/\cosh(V_s/20).$$

In the basal dendrite, the activation gating variable n for I_{KDR} has rate functions $\alpha_n(V_{bd})$ and $\beta_n(V_{bd})$ as follows:

$$\alpha_n(V_{bd}) = \frac{0.00049(V_{bd} - 32)}{1 - \exp(-(V_{bd} - 32)/25)},$$

$$\beta_n(V_{bd}) = \frac{0.00008(V_{bd} - 42)}{\exp((V_{bd} - 42)/10) - 1}.$$

For the NMDA-mediated channel I_{NMDA} , there is a marked voltage dependency in the presence of extracellular Mg^{2+} . For physiological Mg^{2+} concentration, the voltage dependence of I_{NMDA} can be described as follows [19]:

$$B(V_{bd}) = \frac{1}{1 + 0.3 \exp(-0.08V_{bd})}.$$

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